

PRASHANT

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Professional Summary

Computer Science and Engineering student with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, DBMS, and Web Development. Skilled in C++, JavaScript, React.js, Node.js, Express.js, MongoDB, and REST APIs. Experienced in building networking and full-stack applications. Seeking an entry-level Software Development or Software Engineering opportunity.

Education

B.Tech. in Computer Science and Engineering	2023 – 2027
Dr. A.P.J. Abdul Kalam Technical University, Neelkanth Institute of Technology, Meerut	SGPA: 8.61/10
Class XII – Uttar Pradesh Board	2023
Maharishi Dayanand Public Inter College	73.2%
Class X – CBSE	2021
Shatabdi Public School, Meerut	90.05%

Technical Skills

Languages:	C++, Java, Python, JavaScript
Frontend:	HTML, CSS, JavaScript, React.js, Tailwind CSS
Backend:	Node.js, Express.js, REST APIs
Databases:	MySQL, MongoDB
Tools:	Git, GitHub, VS Code, Postman
Core:	Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

Projects

Deep Packet Inspection (DPI) & Packet Analyzer *C++17, Networking, Multithreading*

- Developed a C++17 DPI engine to analyze PCAP network captures and classify application-level network traffic.
- Implemented Ethernet, IPv4, TCP, and UDP packet parsing to extract MAC/IP addresses, ports, protocols, and packet metadata.
- Built five-tuple based flow tracking using hash-based data structures to associate packets with network connections and maintain flow state.
- Implemented TLS SNI and HTTP Host extraction with configurable IP, domain, and application-level blocking rules.
- Designed a multithreaded producer-consumer pipeline using queues, mutexes, condition variables, and worker threads.

URL Shortener *MongoDB, Express.js, Node.js, React.js, REST API*

- Developed a full-stack URL shortening application that converts long URLs into short, shareable links with fast redirection.
- Designed and implemented RESTful APIs for URL creation, validation, management, and redirection using Node.js and Express.js.
- Implemented secure JWT-based user authentication and integrated an Admin Dashboard for managing user links and viewing click analytics.
- Integrated MongoDB for persistent storage of original URLs, shortened URLs, user credentials, and associated metadata.

Certifications

- Generative AI for Everyday Life – SWAYAM (IGNOU & KKHSOU), Score: 96%
- Advanced Software Engineering Job Simulation – Walmart Global Tech (via Forage)
- Basics of Cyber Security – IBM SkillBuild
- Python 3.4.3 Training – Spoken Tutorial, IIT Bombay

Achievements

- Successfully completed Walmart Global Tech's Advanced Software Engineering Job Simulation, implementing advanced data structures, software architecture, and relational database designs.
- Demonstrated proficiency in low-level networking and system development by engineering a custom C++ multi-threaded DPI packet analyzer.
- Solved 250+ data structures and algorithms problems across LeetCode and GeeksforGeeks.
- Maintained a strong academic record with a cumulative SGPA of 8.61/10 in B.Tech Computer Science.
- Contributed as an Open Source Developer in GirlScript Summer of Code (GSSoC).